Supplementary material

Error-bounded reconstruction and loss-resilient delivery of river water-quality records over satellite IoT: A reproducible framework.

Table S1: Development sites (2023): four-parameter record days, median bytes of the record, and the share of days fully certified at six messages per day with independent loss of 0.3.

Site	Name	State	Interval (min)	Days	Median bytes	G_{cert}
01388000	Ramapo River at Pompton Lakes NJ	NJ	15	340	18	0.988
01400500	Raritan River at Manville NJ	NJ	5	79	36	0.919
01402000	Millstone River at Blackwells Mills NJ	NJ	5	218	21	0.974
01427510	Delaware River At Callicoon Ny	PA	15	310	33	0.917
01438500	Delaware River at Montague NJ	PA	30	356	27	0.960
01458500	Delaware River at Frenchtown NJ	NJ	15	328	21	0.984
01463500	Delaware River at Trenton NJ	NJ	60	360	36	0.913
01465798	Poquessing Creek at Grant Ave. at Philadelphia, PA	PA	15	246	56	0.614
014670261	Delaware River at Pennypack Woods PA	PA	15	323	39	0.854
01467087	Frankford Creek at Castor Ave, Philadelphia, PA	PA	15	210	24	0.916
01467200	Delaware River at Penn's Landing, Philadelphia, PA	PA	5	339	27	0.959
01473500	Schuylkill River at Norristown, PA	PA	15	272	39	0.807
01473900	Wissahickon Creek at Fort Washington, PA	PA	30	70	42	0.843
01475530	Cobbs Cr at U.S. Hghwy No. 1 at Philadelphia, PA	PA	15	238	45	0.730
01477050	Delaware River At Chester Pa	PA	15	321	36	0.829
03072655	Monongahela River near Masontown, PA	PA	15	209	27	0.955
03075070	Monongahela River at Elizabeth, PA	PA	15	165	27	0.897
03081000	Youghiogheny River below Confluence, PA	PA	15	168	30	0.933
03083500	Youghiogheny River at Sutersville, PA	PA	15	175	33	0.953
05420400	Mississippi River At Dam 13 Near Fulton, Il	IL	15	343	21	0.983
05537980	Des Plaines River At Route 53 At Joliet, Il	IL	15	294	45	0.730
05543010	Illinois River At Seneca, Il	IL	15	317	18	0.959
05553700	Illinois River At Starved Rock, Il	IL	15	215	21	0.872
05559900	Illinois River Above Rte 150 At Peoria, Il	IL	15	199	24	0.960
05586300	Illinois River At Florence, Il	IL	15	290	18	0.824
11509370	Klamath River Above Keno Dam, At Keno, Or	OR	30	346	27	0.763
11509500	Klamath River At Keno, Or	OR	15	358	18	0.917
11510700	Klamath River Blw John C.Boyle Pwrplnt, Nr Keno,Or	OR	15	359	42	0.860

Site	Name	State	Interval (min)	Days	Median bytes	G_{cert}
14162500	Mckenzie River Near Vida, Or	OR	15	343	27	0.961
14163900	Mckenzie River Near Walterville, Or	OR	15	361	27	0.973
14203500	Tualatin River Near Dilley, Or	OR	15	250	18	0.991
14206241	Tualatin River At Hwy 219 Near Hillsboro, Or	OR	30	348	18	0.994
14206694	Tualatin River At River Mile 24.5, Nr Scholls, Or	OR	30	199	21	0.990
14207200	Tualatin River At Oswego Dam, Near West Linn, Or	OR	30	267	15	0.994
14209710	Clackamas River At Carter Bridge, Near Estacada,Or	OR	30	360	21	0.991
14210000	Clackamas River At Estacada, Or	OR	30	76	12	0.999
14211010	Clackamas River Near Oregon City, Or	OR	15	355	36	0.916
420853121505500	Klamath River At Miller Island Boat Ramp, Or	OR	30	335	24	0.762

Table S2: Confirmation sites (2024), with the same columns as Table S1.

Site	Name	Region	Interval (min)	Days	Median bytes	G_{cert}
01021050	St. Croix River at Milltown, Maine	New England and New York	15	304	18	0.988
01433500	Mongaup River Near Mongaup Ny	New England and New York	15	349	18	0.985
02110704	Waccamaw River At Conway Marina At Conway, Sc	Southeast	15	331	18	0.954
02171645	Rediv Canal At Santee River Nr St Stephen, Sc	Southeast	15	323	27	0.858
02203873	Cobbs Creek At Rainbow Dr, Near Decatur, Ga	Southeast	15	335	30	0.918
02207135	Little Stone Mtn Cr Near Stone Mountain, Ga	Southeast	15	358	24	0.938
02207160	Stone Mountain Creek At Ga 124, Near Lithonia, Ga	Southeast	15	356	21	0.964
02217643	Parks Creek At Lyle Field Rd Nr Jefferson, Ga	Southeast	15	296	27	0.960
02344673	Flat Creek Ds Of Lake At Peachtree City, Ga	Southeast	15	149	21	0.989
02413205	Little Tallapoosa R At Reavesville Rd Nr Bowdon,Ga	Southeast	15	283	24	0.976
04027000	Bad River Near Odanah, Wi	Great Plains and upper Midwest	15	261	21	0.984
04213500	Cattaraugus Creek At Gowanda Ny	New England and New York	15	342	30	0.960
04231600	Genesee River At Ford Street Bridge, Rochester Ny	New England and New York	15	356	21	0.868

Site	Name	Region	Interval (min)	Days	Median bytes	G_{cert}
05545750	Fox River Near New Munster, Wi	Great Plains and upper Midwest	15	266	27	0.980
06711565	South Platte River At Englewood, Co.	Mountain West	15	328	42	0.841
06887500	Kansas R At Wamego, Ks	Great Plains and upper Midwest	15	317	36	0.917
06889110	Soldier Creek Nr Corning, Ks	Great Plains and upper Midwest	15	280	45	0.716
06892350	Kansas R At Desoto, Ks	Great Plains and upper Midwest	15	326	33	0.909
07109500	Arkansas River Near Avondale, Co.	Mountain West	15	286	30	0.953
07143672	L Arkansas R At Hwy 50 Nr Halstead, Ks	Great Plains and upper Midwest	15	250	33	0.905
07144100	L Arkansas R Nr Sedgwick, Ks	Great Plains and upper Midwest	15	328	39	0.860
07332622	Bois D'Arc Ck at FM 409 nr Honey Grove, TX	Southwest and Texas	15	304	42	0.711
08049500	W Fk Trinity Rv at Grand Prairie, TX	Southwest and Texas	15	259	30	0.888
08052700	Little Elm Ck nr Aubrey, TX	Southwest and Texas	15	320	39	0.758
08052745	Doe Br at US Hwy 380 nr Prosper, TX	Southwest and Texas	15	278	30	0.830
08057000	Trinity Rv at Dallas, TX	Southwest and Texas	15	324	27	0.922
08069500	W Fk San Jacinto Rv nr Humble, TX	Southwest and Texas	15	348	45	0.627
08375300	Rio Grande at Rio Grande Village, Big Bnd NP, TX	Southwest and Texas	15	257	39	0.885
09070500	Colorado River Near Dotsero, Co	Mountain West	15	345	27	0.968
09085150	Colorado R Abv South Canyon Cr Nr Glenwood Spgs Co	Mountain West	5	342	30	0.951
09163500	Colorado River Near Colorado-Utah State Line	Mountain West	15	346	21	0.988
09261000	Green River Near Jensen, Ut	Mountain West	15	335	24	0.984
09510000	Verde River Below Bartlett Dam, Az	Southwest and Texas	15	314	27	0.974
10129900	Silver Creek Near Silver Creek Junction, Ut	Mountain West	15	314	42	0.879
375350097262800	L Arkansas R Upstream Of Asr Fac Nr Sedgwick, Ks	Great Plains and upper Midwest	15	325	45	0.778
391953108130201	Roan Creek At De Beque, Co.	Mountain West	15	298	36	0.856

Table S3: Cohort flow: channel-days (one parameter of one site on one UTC day) and record days, development (2023) and confirmation (2024) cohorts.

Stage	Development	Confirmation
Channel-days with data	50,983	50,797
Ineligible (gap of more than two readings, or too few readings)	2,846	2,802
Eligible channel-days	48,137	47,995
Record days (at least one eligible parameter)	12,674	12,380
with one parameter	48	75
with two parameters	131	128
with three parameters	2,153	1,044
with four parameters (primary cohort)	10,342	11,133
Sites with four-parameter days	38	36
Four-parameter days with mixed intervals across parameters	632	6

Table S4: Days fully certified, G_{cert}, on the confirmation sites under all channels (exact; all four-parameter days).

	$Q = 6$				$Q = 12$			
Channel	Code	Rep.	Every k	Tag	Code	Rep.	Every k	Tag
Independent, $p = 0.05$	0.969	0.955	0.118	0.046	0.995	0.991	0.268	0.048
Independent, $p = 0.1$	0.963	0.932	0.086	0.043	0.994	0.985	0.140	0.048
Independent, $p = 0.2$	0.940	0.874	0.042	0.037	0.991	0.971	0.034	0.047
Independent, $p = 0.3$	0.900	0.799	0.019	0.030	0.987	0.948	0.007	0.046
Independent, $p = 0.4$	0.833	0.703	0.008	0.023	0.977	0.910	0.001	0.042
Independent, $p = 0.5$	0.733	0.589	0.003	0.017	0.955	0.848	0.000	0.037
Independent, $p = 0.6$	0.599	0.460	0.001	0.010	0.903	0.750	0.000	0.029
Gilbert, $p = 0.05$, burst 3	0.953	0.950	0.140	0.045	0.993	0.991	0.387	0.047
Gilbert, $p = 0.1$, burst 3	0.930	0.924	0.120	0.042	0.989	0.985	0.294	0.046
Gilbert, $p = 0.2$, burst 3	0.880	0.866	0.083	0.035	0.979	0.969	0.152	0.044
Gilbert, $p = 0.3$, burst 3	0.823	0.798	0.052	0.029	0.966	0.945	0.064	0.042
Gilbert, $p = 0.4$, burst 3	0.758	0.717	0.028	0.023	0.947	0.909	0.019	0.038
Gilbert, $p = 0.5$, burst 3	0.679	0.616	0.011	0.017	0.921	0.852	0.003	0.034
Gilbert, $p = 0.6$, burst 3	0.579	0.483	0.002	0.011	0.879	0.758	0.000	0.028
Gilbert, $p = 0.3$, burst 6	0.764	0.759	0.078	0.030	0.909	0.902	0.153	0.037



Figure S1: Monthly share of days fully certified by the record-first code, six messages per day, independent loss of 0.3 (four-parameter days of each cohort).